

Ameer Sohail Shaik

Data Scientist | AI/ML Engineer | Python, LLMs, ML, Data Engineering
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PROFESSIONAL SUMMARY

M.S. Data Science student with hands-on experience building production ML systems, LLM-powered applications, and automated data pipelines. Shipped a multimodal RAG system across four LLM providers, an AutoML platform reducing model build time by 80%, and an XGBoost classifier reaching 87% accuracy on enterprise SAP data. Strong in Python, SQL, ML, and translating technical work into business outcomes.

KEY SKILLS

Programming & Querying: Python (pandas, NumPy, scikit-learn), SQL (joins, window functions, CTEs), Java, Bash; Git, Agile development

Machine Learning & DL: PyTorch, TensorFlow, scikit-learn; XGBoost, Random Forest, Gradient Boosting; classification, regression, forecasting, clustering, feature engineering, hyperparameter tuning, SHAP interpretability, model evaluation, A/B testing, hypothesis testing

LLMs & Generative AI: OpenAI/GPT, Anthropic Claude, Gemini, Groq APIs; LangChain, RAG pipelines, prompt engineering, ChromaDB, sentence-transformers, hybrid retrieval (dense + BM25), cross-encoder re-ranking

Data Engineering: ETL pipelines (SQL, CSV, JSON, log/text); PySpark, Apache Airflow; schema validation, data quality checks, lineage tracking

Visualization & Tools: Power BI (DAX, Power Query, KPI dashboards), Tableau (basic), Streamlit, matplotlib, seaborn; PostgreSQL, MySQL, SQLite, ChromaDB; cloud architecture (AWS, GCP, Azure – foundational)

PROFESSIONAL EXPERIENCE

Junior Data Scientist | Proceedit | Remote

Aug 2024 – Aug 2025

- Designed an offline evaluation framework in Python comparing 10 ML architectures on directional accuracy, Sharpe ratio, and max draw-down; gradient-boosted models matched deep learning precision (65%) at 12x lower inference cost, blocking an over-engineered LSTM from production.
- Ran feature importance analysis on 8 candidate signals across 200+ tickers, surfacing lagged sector-ETF returns as the strongest predictor and shifting the trading signals team off an ARIMA baseline onto supervised learning.
- Automated a daily data preparation workflow with a Python ETL pipeline (PostgreSQL, Google Sheets API), cutting runtime from 60 minutes to 3 minutes per sync (92% reduction) and freeing 8+ analyst hours per week.
- Wrote production SQL reconciling OHLCV data across 4 source tables and added schema validation that eliminated recurring data quality errors in downstream reporting.

Data Science Intern | SBP Consulting Pvt Ltd | Hyderabad, India

Jan 2024 – May 2024

- Analyzed 12 client accounts' SAP S/4HANA records and engineered recency-weighted purchase frequency and seasonal spend ratios into an XGBoost model achieving 87% accuracy (MAPE 13%); results directly shaped quarterly C-suite planning recommendations.
- Built 5 interactive Power BI dashboards mapping sales trends, churn risk segments, and 7+ KPIs to executive questions, replacing a 2-day manual Excel reporting cycle with a 4-hour automated refresh and cutting ad-hoc data requests by approximately 60%.
- Cut data preparation time by 45% with Python scripts (pandas, NumPy) consolidating 4 SAP modules into analysis-ready formats with standardized null imputation, duplicate detection, and schema validation across 3 downstream pipelines.

PROJECTS

Ragnarok: Multimodal RAG Knowledge Assistant | Python, LangChain, ChromaDB, OpenAI, Claude, Gemini, Groq, Streamlit

- Built a multimodal RAG pipeline ingesting PDFs, audio (Whisper transcription), and video (OpenCV frame extraction with vision-model captions) into a unified ChromaDB vector store of 10K+ chunks across 4 modalities; combined dense cosine search, BM25 sparse ranking via Reciprocal Rank Fusion, and cross-encoder re-ranking (ms-marco-MiniLM) to reduce irrelevant context passed to the generation layer by approximately 60%.
- Routed queries across 4 LLM providers (Groq, OpenAI, Gemini, Claude) behind a citation-enforcing system prompt, deployed via Streamlit with production caching, async batch embedding, and configurable retrieval parameters.

AutoML-ify: No-Code AutoML Platform | Python, Streamlit, Scikit-learn

- Built an end-to-end AutoML web application enabling non-technical users to train ML models without code, supporting 5+ algorithms with automated hyperparameter tuning via GridSearchCV across datasets up to 100K rows; designed an evaluation dashboard with model metrics and visualizations that cut model development time by 80% for business analyst users.

Diabetes Risk Analysis & Prediction | Python, XGBoost, Random Forest, SHAP

- Built a binary classification system predicting diabetes risk from 21 health indicators using ensemble methods (XGBoost, Random Forest, Gradient Boosting) on 150K+ CDC BRFSS records, reaching 95.9% recall and 0.9935 ROC-AUC; engineered 18 features (interaction terms, composite health scores) improving recall by 12% and applied SHAP to identify GenHlth, BMI, and Age as top risk factors.

Dynamic SQL Assistant: Text-to-SQL Engine | Python, LangChain, Groq, SQLite

- Built a Text-to-SQL engine enabling natural language database queries with 90%+ semantic accuracy across 50+ tested patterns; designed prompt orchestration with runtime schema injection and SQL validation to guarantee syntactic correctness.

EDUCATION

University of Maryland, College Park | M.S. in Data Science

Aug 2025 – Expected May 2027

Vellore Institute of Technology, AP | B.Tech in Computer Science and Engineering

2020 – 2024